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Seasonal interactive effects of pCO₂ and irradiance on the ecophysiology of brown macroalga *Fucus vesiculosus* L.

Antti Takolander ^{a,b}, Mar Cabeza^a and Elina Leskinen^c

^aGlobal Change and Conservation Group, Helsinki Institute of Sustainability Science, Faculty of Biological and Environmental Sciences, University of Helsinki, PO Box 65 (Viikinkaari 1) FI-00014 Helsinki, Finland; ^bTvärminne Zoological Station, University of Helsinki, J. A. Palménin tie 260, FI 10900 Hanko, Finland; ^cEcosystems and Environment Research Programme, Faculty of Biological and Environmental Sciences, University of Helsinki, PO Box 65 (Viikinkaari 1), FI 00014 Helsinki, Finland

ABSTRACT

Stochastic upwelling of seawater in the Baltic Sea from the deep, anoxic bottoms may bring low-pH water rich in CO₂ close to the surface. Such events may become more frequent with climate change and ongoing ocean acidification (OA). Photoautotrophs, such as macroalgae, which are important foundation species, have been proposed to benefit from increased carbon availability due to reduced energetic cost in carbon acquisition. However, the exact effects of CO₂ fertilization may depend on the ambient light environment, as photosynthesis rates depend on available irradiance. In this experimental study, interacting effects of CO₂ addition and irradiance on the habitat-forming macroalga *Fucus vesiculosus* were investigated during two seasons – winter and summer – in the northern Baltic Sea. Growth rates remained unaffected by CO₂ or irradiance during both seasons, suggesting that direct effects of elevated CO₂ on mature *F. vesiculosus* are small. Increases in CO₂ affected algal elemental ratios by increasing carbon and decreasing nitrogen content, with resulting changes in the C:N ratio, but only in winter. In summer, chlorophyll *a* content increased under low irradiance. Increases in CO₂ caused a decline in light-harvesting efficiency (decrease in F_v/F_m and α) under high irradiance in summer, and conversely increased α under low irradiance. High irradiance caused increases in the maximum relative electron transport rate ($rETR_{max}$) in summer, but not in winter. Differences between winter and summer indicate that *F. vesiculosus* responses to CO₂ and irradiance are season-specific. Increases in carbon content during winter could indicate slightly positive effects of CO₂ addition in the long run if the extra carbon gained may be capitalized in growth. The results of this study suggest that increases in CO₂, either through upwelling or OA, may have positive effects on *F. vesiculosus*, but these effects are probably small.

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Introduction

The Baltic Sea is one of the world's largest brackish water pools, characterized by low salinity and low dissolved inorganic carbon (DIC) content compared with ocean areas (Thomas & Schneider, 1999; Snoeijs-Leijonmalm *et al.*, 2017). The Baltic Sea has suffered heavily from anthropogenic eutrophication, which has caused hypoxia and anoxia in deep water, and subsequently led to the occurrence of spreading dead zones on the seafloor (Conley *et al.*, 2011; Carstensen *et al.*, 2014; van Helmond *et al.*, 2018). Seawater upwelling from these areas may cause water low in O₂ and rich in CO₂ to also affect coastal littoral communities (Saderne *et al.*, 2013). Coastal eutrophication has also caused hypoxic conditions in shallow coastal areas, as drifting macroalgal mats decompose, causing declines in O₂, elevated pCO₂ and declining pH (Bonsdorff *et al.*, 1997; Sunda & Cai, 2012).

Coastal ecosystems have highly variable ambient CO₂ concentrations caused by seasonal changes in

photosynthesis, respiration and upwelling (Thomas & Schneider, 1999; Borges *et al.*, 2006; Omstedt *et al.*, 2009). Future seasonal pH fluctuations in the Baltic Sea are expected to increase in magnitude under the combined effects of eutrophication and climate change, with concomitant decreases in mean pH (Omstedt *et al.*, 2012). Ocean acidification (OA) may be especially rapid in the Baltic Sea, as cold water readily absorbs CO₂, and the northern parts in particular have low water alkalinity (Müller *et al.*, 2016), which intensifies the expected pH decreases by OA (Omstedt *et al.*, 2010). Declining sea ice extent may concurrently increase light availability during, for example, low pH periods in winter. On the other hand, intensifying eutrophication owing to climate change (Meier *et al.*, 2012; Neumann *et al.*, 2012) may cause diminished underwater light conditions. Hypoxia, a major consequence of eutrophication in the Baltic, may significantly enhance OA impacts, and stochastic seawater pulses with extremely high CO₂, originating from hypoxic bottom areas, may affect Baltic coastal biotic communities in the future

(Melnzer *et al.*, 2013). Thus both the underwater light regime and CO₂ availability are expected to change in the future Baltic.

The DIC pool of seawater consists of three different fractions: carbonic acid (H₂CO₃); bicarbonate (HCO₃⁻); and carbonate (CO₃²⁻) (Fabry *et al.*, 2008). Although seawater has a greater supply of inorganic carbon than air, the majority of this DIC pool is in the form of HCO₃⁻, which needs to be converted to CO₂ before it can be utilized by the Rubisco of primary producers as a carbon source (Kirk, 2011).

CO₂ diffusion in seawater is an order of magnitude slower than in air (Falkowski & Raven, 2007), meaning that when photosynthesis rates are high, carbon may be depleted in the immediate proximity of a photoautotroph. In highly productive shallow water habitats, free CO₂ may also be scarce, as it is taken up for photosynthesis (Middelboe & Hansen, 2007). To avoid potential carbon limitation, many primary producers, such as macroalgae, have evolved carbon concentrating mechanisms (CCMs) that increase free CO₂ concentrations near the Rubisco binding site, allowing organisms to also utilize HCO₃⁻ as a carbon source (Raven & Hurd, 2012), with the energetic cost associated with production and upkeep of CCMs (Raven *et al.*, 2014).

Despite possessing CCMs, the photosynthesis of most macroalgal species studied does not appear to be saturated at present levels of inorganic carbon in the seawater (Koch *et al.*, 2013). Climate change will increase CO₂ availability in seawater, as atmospheric dissolution of CO₂ increases (Orr *et al.*, 2005). This has been suggested to confer energetic advantages to primary producers, as they could potentially downregulate CCM utilization (Koch *et al.*, 2013). Increasing diffusive CO₂ usage as a carbon source would be more energy efficient in comparison to relying on active HCO₃⁻ transport as a carbon source for photosynthesis (Cornwall *et al.*, 2012).

Different macroalgal species show variable responses to increased CO₂ availability, studied mainly in the context of OA. These differential responses may be caused by an interplay between various environmental factors and variable species-specific traits in carbon metabolism (Cornwall *et al.*, 2012). Species with active CCMs, often residing under high irradiance, may benefit from CO₂ fertilization (Mercado & Gordillo, 2011), as an energy-consuming CCM may be downregulated under increased CO₂ concentrations, allowing improved carbon energetics (Raven *et al.*, 2011). On the other hand, increased CO₂ may lower tolerance of high light intensities (Liu *et al.*, 2012), as the CCM acts as a sink for excessive energy (Wu *et al.*, 2008). Indeed, marine primary production has been predicted to decrease under the coupled high irradiance and high CO₂

predicted for the future, as primary producers downregulate photosynthetic machinery (Gao *et al.*, 2012). It is thus likely that the exact effects of CO₂ availability on macroalgal physiology depend on irradiance (Verspagen *et al.*, 2014; Celis-Plá *et al.*, 2015; Kübler & Dudgeon, 2015).

Fucus vesiculosus L. is the major habitat-forming macroalga in the northern Baltic Sea (Waern 1952; Kautsky *et al.*, 1992), harbouring a rich community of associated floral and faunal species (Schagerström *et al.*, 2014). In the Baltic, *F. vesiculosus* has suffered due to eutrophication, and may be threatened by climate change (Vuorinen *et al.*, 2015; Takolander *et al.*, 2017a; Rugiu *et al.*, 2018). *Fucus vesiculosus* has a highly efficient CCM (Giordano & Maberly, 1989; Surif & Raven, 1989); nevertheless, photosynthesis of the Baltic Sea populations is carbon limited due to the low carbon content of brackish water (Raven & Samuelsson, 1988). Permanently submerged Baltic populations undergo substantial fluctuations in environmental conditions in different seasons, and the ecophysiology of *F. vesiculosus* also shows seasonal variability, with plants storing nitrogen and mannitol (carbon sink), which can be utilized under times of diminished external supply (Lehvo *et al.*, 2001). Studies on OA effects (CO₂ fertilization) on *F. vesiculosus* have yielded mixed results, showing reduced growth (Gutow *et al.*, 2014), no effect (Pajusalu *et al.*, 2013) or increased juvenile survival in particular seasons (Al-Janabi *et al.*, 2016b), reviewed in detail in Takolander *et al.* (2017b).

Interacting drivers may have synergistic effects (Wahl *et al.*, 2011), and quantifying such responses for foundation species has high ecological importance. As the ecophysiology of *F. vesiculosus* shows seasonal variations, the responses to such interactions may differ by season (Al-Janabi *et al.*, 2016b; Werner *et al.*, 2016). OA effects on *F. vesiculosus* have been investigated in earlier studies, but the interactive effects of light and CO₂ remain unexplored, although these might explain some of the contrasting responses which have been observed. In this study, we investigated the ecophysiological responses of *F. vesiculosus* to the interaction of two environmental variables which are expected to change in the future (light and CO₂) during two seasons (winter and summer) in the northern Baltic Sea. We hypothesized that the effects of increasing CO₂ availability would have positive effects for *F. vesiculosus*, but only under low irradiance.

Methods

Experimental set-up and seawater parameters

The experiment was performed in two seasons – winter and summer – at Tvärminne Zoological

Station (TZS), SW Finland. The winter experiment was conducted in November/December 2015 and the summer experiment in June 2016. The duration of the experiments was 18 days in winter and 22 days in summer.

The algae were collected with a rake or by snorkelling from two islands near TZS – Brännskär and Granbusken. Only vegetative tips free of epiphytes were sampled. The individuals collected were kept fully submerged and were rapidly transported to the laboratory, where they were placed in constant seawater flow-through for 4 days until the onset of the experiments with light conditions and light/dark rhythm as described later in this section. The experimental treatments consisted of two factors: CO₂ treatment (three pCO₂ levels) and light treatment (two levels).

Fucus vesiculosus fronds (all fronds lacked vesicles, and thus they remained at the bottom of the jars) were placed in 1 l glass jars. The jars were placed under two light regimes: 'high' and 'low' irradiance, provided by Philips TL-D Super 80 58W 830 fluorescent light bulbs, with 8:16 h (winter) and 16:8 h (summer) light/dark rhythm. Irradiance was monitored with HOBO Pendant UA-002-64 temperature and light loggers (Onset Computer Corporation), which were calibrated against a factory-calibrated light sensor (LI-COR LI-1500) using the exponential decay fit function suggested by Long *et al.* (2012). Irradiances measured at the water surface of the jars were 165 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ and 81 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (winter) and 198 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ and 131 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (summer) for high and low light treatments, respectively. These light levels correspond approximately to field light conditions at 3 m depth in June and at 0.5 m depth in January (Lindström, 2000).

In the winter experiment, a single frond was placed in one jar, and in the summer experiment, three marked individuals were placed in each jar. Each light treatment contained 30 jars, with 10

replicate jars for each CO₂ treatment level. Mean algal biomass per jar at the beginning of the experiment was 1.02 ± 0.3 g in winter and 2.35 ± 0.7 g in summer.

CO₂ treatments were administered by adjusting pH through bubbling gaseous CO₂ into seawater with Aqua Medic pH controllers (AB Aqua Medic GmbH). The adjustment was made in header tanks (volume = 100 l) into which seawater was pumped from the nearby bay, from a depth of 10 m. The seawater was run through a series of filters (100, 50 and 25 μm) prior to being used in the experiment. One header tank was used for each CO₂ treatment. The bottom of each jar containing *F. vesiculosus* fronds received 80 ml min⁻¹ flow-through of pH-adjusted water from a header tank through a 4 mm diameter silicon hose, which provided water motion, and replenished the water approximately every 12 min. At the beginning of the experiment, the pH levels were adjusted slowly over 24 h to avoid shock effects. Throughout the experiment, the pH levels in all jars were monitored by measuring them every second or third day (11 times in 22 days; Supplementary fig. S2) with a handheld pH meter (pH 1000 H, VWR, with pHenomenal 111 probe, 0.01 units accuracy and precision). Measurement accuracy was checked regularly against a bench-top pH meter (Jenway 3510, with Jenway 924030 Tris electrode). The bench-top pH meter has been annually intercalibrated in Proficiency tests by the Finnish Environmental Institute (Leivuori *et al.*, 2018) using natural water samples. Both pH probes used in the experiment were calibrated with commercial NIST-traceable buffers (Merck Certipur®) using 3-point (pH 4, 7 and 10) calibration. The OA treatments had three pCO₂ levels: 'ambient' (236 μatm or 512 μatm , summer/winter), 'high' (1582 μatm or 2263 μatm , summer/winter) and 'extreme' (4673 μatm or 7074 μatm , summer/winter) (Table 1, values are means across the two light treatments, calculated from all measurements for the duration of the

Table 1. Seawater parameters during the two experiments.

Season	Winter			Summer		
pCO ₂ treatment	Ambient	High	Extreme	Ambient	High	Extreme
pH	7.97 $\pm$ 0.07	7.34 $\pm$ 0.03	6.84 $\pm$ 0.02	8.22 $\pm$ 0.19	7.36 $\pm$ 0.16	6.99 $\pm$ 0.13
Salinity	6.0 $\pm$ 0.1	6.0 $\pm$ 0.1	6.0 $\pm$ 0.1	5.09 $\pm$ 0.14	5.09 $\pm$ 0.15	5.09 $\pm$ 0.14
Temperature (°C)	8.88 $\pm$ 0.28	9.06 $\pm$ 0.33	9.07 $\pm$ 0.33	13.49 $\pm$ 0.85	13.54 $\pm$ 1.02	13.62 $\pm$ 0.83
NH ₄ ⁺ ($\mu\text{g l}^{-1}$)	3.1 $\pm$ 1.0	3.1 $\pm$ 1.0	3.1 $\pm$ 1.0	8.5 $\pm$ 2.2	8.5 $\pm$ 2.2	8.5 $\pm$ 2.2
NO ₂ ⁻ + NO ₃ ⁻ ($\mu\text{g l}^{-1}$)	85.9 $\pm$ 12.3	85.9 $\pm$ 12.3	85.9 $\pm$ 12.3	26.3 $\pm$ 7.5	26.3 $\pm$ 7.5	26.3 $\pm$ 7.5
PO ₄ ³⁻ ($\mu\text{g l}^{-1}$)	21.7 $\pm$ 1.0	21.7 $\pm$ 1.0	21.7 $\pm$ 1.0	11.9 $\pm$ 1.3	11.9 $\pm$ 1.3	11.9 $\pm$ 1.3
Alkalinity ($\mu\text{mol kg}^{-1}$)	1653 $\pm$ 42	1640 $\pm$ 42	1613 $\pm$ 50	1584 $\pm$ 49	1584 $\pm$ 35	1610 $\pm$ 39
DIC ($\mu\text{mol kg}^{-1}$)	1644 $\pm$ 37	1752 $\pm$ 44	1986 $\pm$ 69	1523 $\pm$ 47	1647 $\pm$ 60	1828 $\pm$ 44
HCO ₃ ⁻ ($\mu\text{mol kg}^{-1}$)	1586 $\pm$ 35	1624 $\pm$ 41	1609 $\pm$ 50	1453 $\pm$ 45	1562 $\pm$ 34	1602 $\pm$ 39
CO ₂ ($\mu\text{mol kg}^{-1}$)	27 $\pm$ 2	121 $\pm$ 2	375 $\pm$ 19	11	75 $\pm$ 2	223 $\pm$ 5
CO ₃ ²⁻ ($\mu\text{mol kg}^{-1}$)	31 $\pm$ 4	7	2	60 $\pm$ 2	10	4
pCO ₂ (μatm)	510 $\pm$ 50	2260 $\pm$ 41	7039 $\pm$ 296	236 $\pm$ 7	1582 $\pm$ 35	4681 $\pm$ 113

Note: Values are means across the duration of the experiment (i.e. one mean value was calculated from all daily measurements) $\pm$ standard deviation. For NH₄⁺, NO₂⁻ & NO₃⁻ and PO₄³⁻ values are the same across all pCO₂ treatments as they were measured from the inflowing seawater.

experiment). The ‘ambient’ treatment consisted of unaltered filtered flow-through seawater.

The pH levels in the ‘ambient’ and ‘high’ treatments (Table 1) lie within the current observed variability of pH levels (1963–2016 mean 8.03, minimum 7.07, maximum 8.95) in the Gulf of Finland open sea area (ICES, 2014; Supplementary fig. S7). Although the CO₂ treatment levels in the ‘high’ and ‘extreme’ treatment are high compared with annual mean pCO₂ seawater values (Supplementary fig. S5), stochastic upwelling has been observed to already have exposed coastal macrophyte habitats to pCO₂ > 2500 µatm in a eutrophied fjord in Denmark (Saderne *et al.*, 2013) with salinity of 20, thus having higher alkalinity levels than the northern Baltic. Recent simulations in salinities of 20 and 35 predict peak pCO₂ levels of 4500 µatm and 3400 µatm (temperature = 10°C) with a doubling of sea surface CO₂ concentration, when oxygen is depleted by heterotrophic respiration (Melzner *et al.*, 2013). Eutrophication is expected to increase concurrently with climate change in the Baltic (Meier *et al.*, 2012; Neumann *et al.*, 2012), which leads to increased pH oscillations due to increased respiration and photosynthesis (Omstedt *et al.*, 2012), and declining oxygen conditions, which intensifies acidification (Burnett, 1997; Melzner *et al.*, 2013). Finally, strong upwelling events are frequent in the study area (Haapala, 1994), which supports the relevance of also examining the effects of extreme CO₂ scenarios on coastal biota (Melzner *et al.*, 2013). Macronutrients (NH₄⁺, NO₂⁻, NO₃⁻ and PO₄³⁻) in the water flowing into the header tanks were monitored throughout the experiment. As the flow-through rate was high and algal biomass in each jar was low, we decided to measure the nutrient levels from the inflowing water. Nutrient analyses were performed by TZS’s laboratory, according to Koroleff (1983a, 1983b).

Temperature and salinity were measured every 2–3 days with a conductivity meter (EC-300, VWR). Salinity and temperature followed fluctuations in ambient seawater. Salinity increased slightly during both experiments (from 6.0 to 6.2 and from 4.9 to 5.2, winter/summer, Supplementary fig. S1a, b), whereas temperature decreased slightly (from 9.7 to 9.0°C and from 14.7 to 13.4°C, winter/summer, Supplementary fig. S1c, d). In addition, temperature and light were monitored every 15 min with HOBO temperature/light loggers.

The winter and summer seasons showed differences when comparing the seawater parameters, most notably salinity, temperature and ambient pH (Table 1, Supplementary figs S1, S2). Due to complications with balancing the flow-through system, pH values in the ‘extreme’ pCO₂ treatment deviated somewhat between summer and winter (Table 1, Supplementary fig. S2). Differences between summer

and winter in the ‘ambient’ treatment were caused by natural seasonal fluctuations in the pH of the seawater.

Total DIC was measured twice during the winter experiment (days 4 and 14), and once during the summer experiment (day 12). In the winter experiment, three replicate samples were collected from each jar, while five replicate samples were collected from each header tank in the summer experiment. The seawater for the alkalinity and DIC measurements was collected with a syringe into glass bottles with airtight seals, which had been rinsed with ion-exchanged water.

DIC content of the samples was measured with a carbon analyser (Elektro-Dynamo URAS-3E). According to the method developed by Salonen (1981), 0.3 ml of the DIC sample was injected into the bubbling chamber of the analyser, converted to CO₂ by continuous addition of acid, and led to an infrared gas analyser using pure gaseous nitrogen as the carrier gas and NaHCO₃ solution as the standard.

To check the accuracy of the DIC measurements, total alkalinity was measured twice during the winter experiment by titrating 50 ml seawater to pH 4.5 with 20 mM HCl (SFS 3005, 1981). This titration corresponds to ISO standard NS 4754: ISO 9963-1:1994 and has 2 µmol l⁻¹ accuracy. The alkalinity samples were allowed to equilibrate to room temperature in darkness. After this, 50 ml of sample were transferred with a glass pipette into a 100 ml open glass beaker with a magnetic stirrer. Twenty mM HCl was injected into the beaker with a table-top titrator (Schott Titronic Basic), and the volume of HCl was recorded with 0.01 ml precision. pH was monitored with a Jenway 3510 pH meter.

Seawater carbon chemistry (alkalinity and partitioning of DIC into CO₂, CO₃²⁻ and HCO₃⁻) was calculated from temperature, salinity, DIC and pH data using the R package ‘seacarb’ (Gattuso *et al.*, 2015). The alkalinity values calculated in this way were compared against titrated alkalinity values and were accurate (mean difference 3 µmol kg⁻¹; Supplementary fig. S3).

Growth rate, carbon and nitrogen content

Fresh weight (FW) was measured with mg precision. Prior to the measurements, individuals were carefully blotted with a paper cloth to remove any extra water and were kept constantly submerged prior to and after the measurement. Relative growth rate (RGR) was calculated from FW according to Lüning (1990), referred to in Olischläger *et al.* (2013) by the equation

$$RGR = \frac{100 * \ln(N_t \div N_0)}{t}, \quad (1)$$

where RGR is the daily growth (% FW), N_t is the FW at day t , N_0 is the initial FW, and t is the time interval in days.

Carbon and nitrogen content (% dry weight, DW) was analysed with a LECO TruSpec Micro CHNS analyser from dried (24 h 60°C) samples. After drying, the samples were ground with a mortar and pestle.

Chlorophyll content

For chlorophyll a quantification, a small piece (29 ± 12 mg FW) of frozen algae was ground with a mortar and pestle in darkness. Chlorophyll was extracted in the dark over 4 h with 10 ml of 94% ethanol. Absorbance spectra of the extract were analysed with a Shimadzu UV-2550 Spectrophotometer with 1-nm precision. Chlorophyll a content was calculated from absorption spectra with equations from Ritchie (2008). Chlorophyll content (in mg) was normalized to algal fresh weight (g).

Fluorescence measurements

Several chlorophyll fluorescence parameters were measured to quantify the responses of photosynthetic machinery to the treatments. All fluorescence measurements were conducted using a Diving-PAM pulse-amplitude modulated fluorometer (Walz GmbH). Maximum quantum efficiency of photosystem II (PS II) photochemistry, F_v/F_m , was measured at the end of the experiment after 12 h of dark adaptation, to ensure that all non-photochemical quenching components were relaxed (Maxwell & Johnson, 2000). Measurements were made by first attaching a dark leaf clip onto the thallus tip, then attaching the fibre optics of the Diving-PAM to the clip to maintain a steady distance between the fibre optics and the clip. After this, F_0 was measured with a short burst (0.2 s) of 680 nm measuring light, followed by a saturating ($> 10\,000 \mu\text{mol m}^{-2} \text{s}^{-1}$) light burst to saturate photosynthesis, and F_m was measured. F_v/F_m was calculated with the equation

$$F_v/F_m = \frac{F_m - F_0}{F_m} \quad (2)$$

The Diving-PAM measuring light intensity and the distance between the sample and the tip of the fibreoptics were adjusted so that the F_0 reading fell between 300 and 500 (arbitrary units, as recommended by the manufacturer). The efficiency of light-limited photosynthetic energy capture (α), maximum relative electron transport rate through PS II ($rETR_{\text{max}}$) and onset of light saturation of photosynthesis (E_k) were measured with rapid light curve (RLC) protocol (Ralph & Gademann, 2005). The halogen lamp of the Diving-PAM was used as an

actinic light source. The internal light library of the Diving-PAM was calibrated against a factory-calibrated light sensor (LI-COR LI-1500). Light intensity was increased from $0 \mu\text{mol m}^{-2} \text{s}^{-1}$ to $681 \mu\text{mol m}^{-2} \text{s}^{-1}$ in eight 20 s increments. As we did not measure absorbance of the tips, we give ETR values as relative electron transport rate ($rETR$) (Ralph & Gademann, 2005). $rETR$ in each light intensity was calculated by the equation

$$rETR = \frac{\Delta F}{F_m'} * PAR * 0.5, \quad (3)$$

where $\Delta F/F_m'$ is the effective quantum yield under actinic light, PAR is the photon flux density of photosynthetically active radiation and 0.5 is the factor accounting for assumed equal partitioning of photons between photosystems I and II (Genty *et al.*, 1989). $rETR$ versus PAR RLCs were fitted to a model by Platt *et al.* (1981) with R package 'phytotools' (Silsbe & Malkin, 2015) and fit method 'PORT', and α , $rETR_{\text{max}}$ and E_k were solved from the equation.

Fresh weight was measured at the beginning and end of the experiments. All fluorescence measurements were taken at the end of the experiment, after which each specimen was split into three Eppendorf tubes and frozen (-80°C) for carbon, nitrogen and chlorophyll analysis.

Statistical analyses

For the summer experiment, the effects of $p\text{CO}_2$ treatment and light on all variables measured were analysed with generalized mixed models using the 'nlme' package (Pinheiro *et al.*, 2015) in R. As three individuals were kept in a single jar, jar ID was included as a random intercept in all the models. For the winter experiment, an ANOVA fit with generalized least squares (GLS) was applied. $p\text{CO}_2$ and light were analysed as factorial covariates with three or two levels, respectively. We followed a backwards-stepwise model selection strategy suggested by Zuur *et al.* (2009), starting with a full model with $p\text{CO}_2$ and light interaction and random intercept for the jar ID (summer experiment). Models were fit with GLS using restricted maximum likelihood estimation (REML). If heterogeneity was observed when plotting model residuals against covariates, variance structures allowing for heterogeneity (Zuur *et al.*, 2009) were applied (Supplementary table S1), and/or the response variable was log-transformed (growth rate and E_k , summer and chlorophyll c content, winter). After this, the significance of the fixed terms was analysed by fitting a new model with GLS using maximum likelihood (ML) and comparing nested models using likelihood ratio tests. We used a P -value of 0.05 as the threshold for accepting a variable into the model. A similar model selection protocol

was applied for the winter and summer data, with the exception of random factors in the summer experiment. When analysing F_v/F_m results in the summer experiment, jar horizontal position within the light treatment was also included as a second random effect in the model (see Supplementary Information section 3.2 for full explanation). After all the significant terms were identified, validity of the model assumptions was checked by visually comparing the normalized residuals against fitted values and all covariates.

To investigate the magnitude of seasonal versus treatment effects, we examined the standardized regression coefficients for the main effects of season, light and pCO_2 on all measured variables. Because the measured variables were on different scales, all response variables were standardized to have a mean of 0 and a standard deviation of 1. For this analysis, the three summer measurements within a single jar were pooled. Absolute values of standardized main effects of the factor season (time of experiment) were compared with those of the light treatment (one level, Fig. 4) and the two pCO_2 treatments (two levels, Fig. 4) using Welch's ANOVA, followed by Games-Howell *post hoc* tests, as the standardized regression coefficients of the various main effects examined had heterogeneous variances. All numerical analyses were conducted in the R software environment version 3.2.0 (R Core Team, 2015).

Results

Growth rate, carbon and nitrogen content

Neither pCO_2 nor light significantly affected growth rate. pCO_2 and light had similar effects on the carbon and nitrogen content and carbon:nitrogen (C:N) ratio of the algae, and the observed effects showed differences between the two seasons investigated (Table 2, Fig. 1). Increasing pCO_2 and light significantly increased carbon content and the C:N ratio, and caused declines in nitrogen content, but, with the

exception of carbon content, these were only observed during winter. The significant effect of pCO_2 on carbon content in summer was rather weak, with 'high' treatment having 0.20% lower and 'extreme' treatment 0.46% higher carbon content than the 'ambient' treatment.

Nitrogen content was not affected by pCO_2 or light treatments in the summer, but increasing pCO_2 and light in winter caused significant declines in nitrogen content (Fig. 1C, Table 2). Likewise, the C:N ratio was unaffected by treatments during summer, but during winter increased light and CO_2 availability caused elevated C:N ratios (Fig. 1D, Table 2).

Chlorophyll content

Chlorophyll *a* content was relatively unaffected by pCO_2 and light, with only significant effects observed in summer, when low light caused higher chlorophyll content (Fig. 2, Table 3). This effect was not observed in winter, but in winter the chlorophyll content was higher overall compared with summer (Fig. 2).

Fluorescence parameters

The mixed model indicated significant interaction between pCO_2 and light on maximum potential quantum yield (F_v/F_m) during summer (Fig. 3A, Table 4). Elevated pCO_2 caused F_v/F_m to decline, but only under high light treatment (Fig. 3A). pCO_2 and light both had significant effects on F_v/F_m during winter (Table 4), with 'extreme' pCO_2 treatment causing decreases in F_v/F_m , and individuals subjected to high light treatment having lower F_v/F_m values (Fig. 3A).

Responses in α showed similar patterns, declining during the summer with increasing pCO_2 , but only under high irradiance (Fig. 3B). Although pCO_2 and light interaction was non-significant ($p = 0.059$), dropping the interaction caused residual patterns (Supplementary fig. S11), which indicate that the interaction probably should have been included in the final

Table 2. Statistical analysis (Mixed model or ANOVA) results for growth rate, carbon and nitrogen content and C:N ratio measurements.

	Growth rate		Carbon content (% DW)		Nitrogen content (% DW)		C:N ratio	
	L	p	F	p	F	p	F	p
Summer								
pCO_2 *Light	1.902	0.386	2.202	0.113	0.834	0.435	0.948	0.389
pCO_2	1.638	0.440	4.596	0.011*	2.052	0.131	2.074	0.128
Light	0.214	0.643	2.947	0.087	2.074	0.128	0.688	0.408
Winter								
pCO_2 *Light	0.958	0.389	0.435	0.649	0.389	0.679	1.906	0.385
pCO_2	0.182	0.833	7.537	0.001**	10.247	< 0.001***	22.007	< 0.001***
Light	1.538	0.219	74.52	< 0.001***	14.352	< 0.001***	27.309	< 0.001***

Note: DW = dry weight. Statistically significant ($p < 0.05$) effects are indicated in bold and with asterisks. Significance levels: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. L indicates likelihood ratio (mixed model), and F an F-statistic (ANOVA), depending on the conducted analysis. Degrees of freedom are the number of treatment levels (three for CO_2 and two for light) - 1.

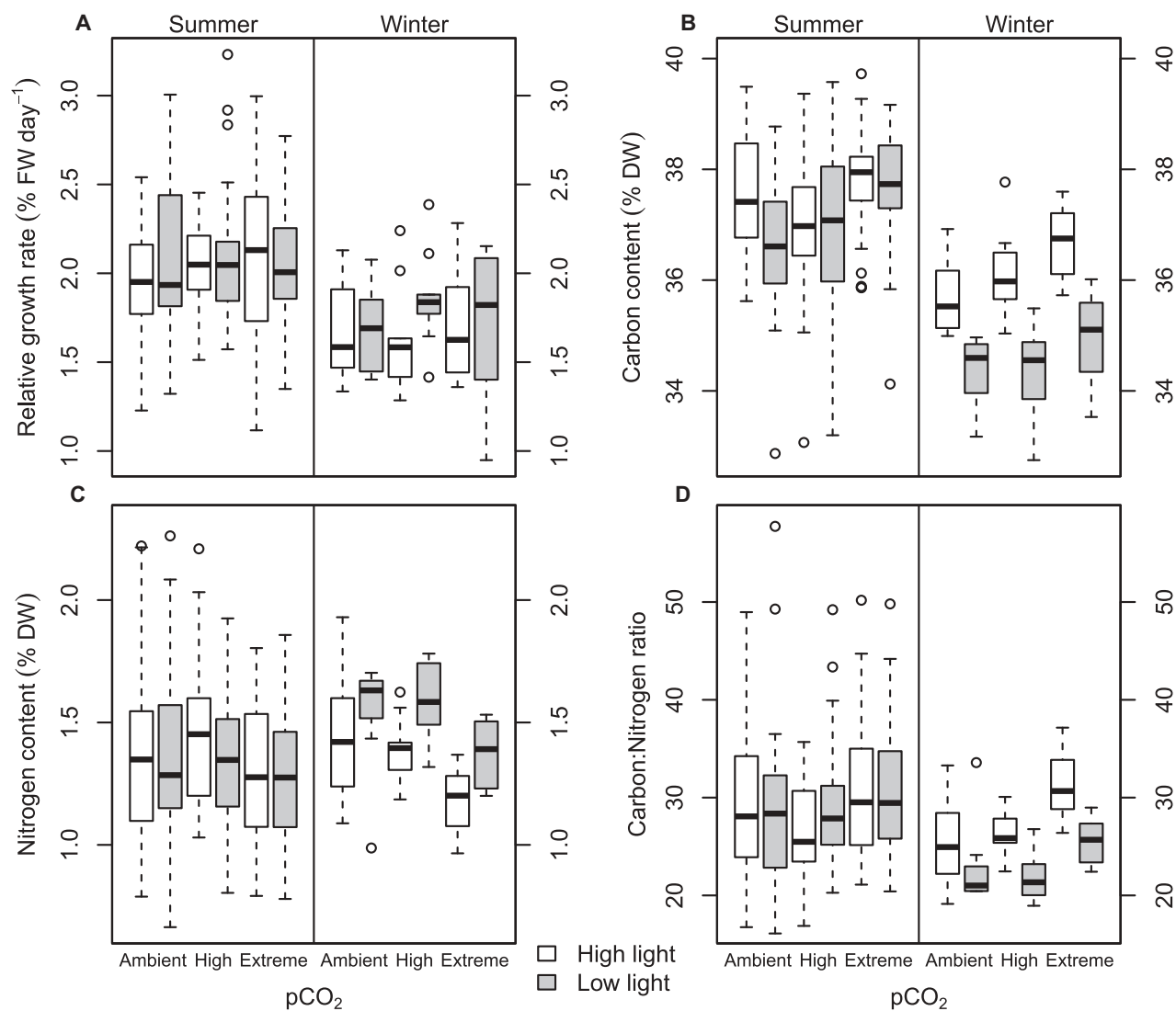


Fig. 1. Effects of pCO₂, light and season on: growth rate (A), carbon content (B), nitrogen content (C) and carbon:nitrogen ratio (D) of *Fucus vesiculosus*.

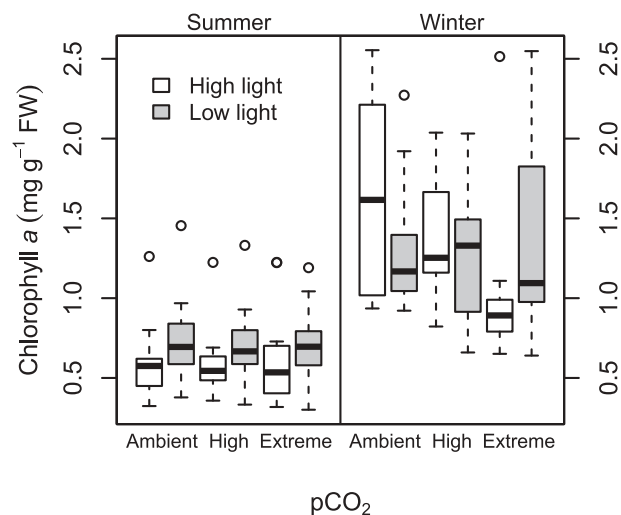


Fig. 2. Effects of pCO₂, light and season on chlorophyll *a* content of *Fucus vesiculosus*.

model. Similar to the F_v/F_m values, individuals exposed to high irradiance during winter had lower values of α , but pCO₂ had no effect (Table 4, Fig. 3B).

Table 3. Statistical analysis (ANOVA) results for chlorophyll *a* content.

	Chlorophyll <i>a</i>			
	Summer		Winter	
	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>
pCO ₂ *Light	0.099	0.905	1.825	0.170
pCO ₂	0.031	0.968	1.834	0.169
Light	9.811	0.002**	0.128	0.721

Note: Statistically significant ($p < 0.05$) effects are indicated by bold and asterisks. Significance level: **, $p < 0.01$. *F* is an *F*-statistic (ANOVA). Degrees of freedom are the number of treatment levels (3 for CO₂ and 2 for light) – 1.

High light treatment caused increases in maximum electron transport rate during the summer, but not in winter. pCO₂ treatment had no effect on rETR_{max} in summer, but high pCO₂ caused decreases in rETR_{max} in winter (Table 4, Fig. 3C).

Fucus vesiculosus plants acclimated to irradiance treatments during the experiment, which was seen in high light-treated plants having higher compensation points for photosynthesis (E_k) in both winter and

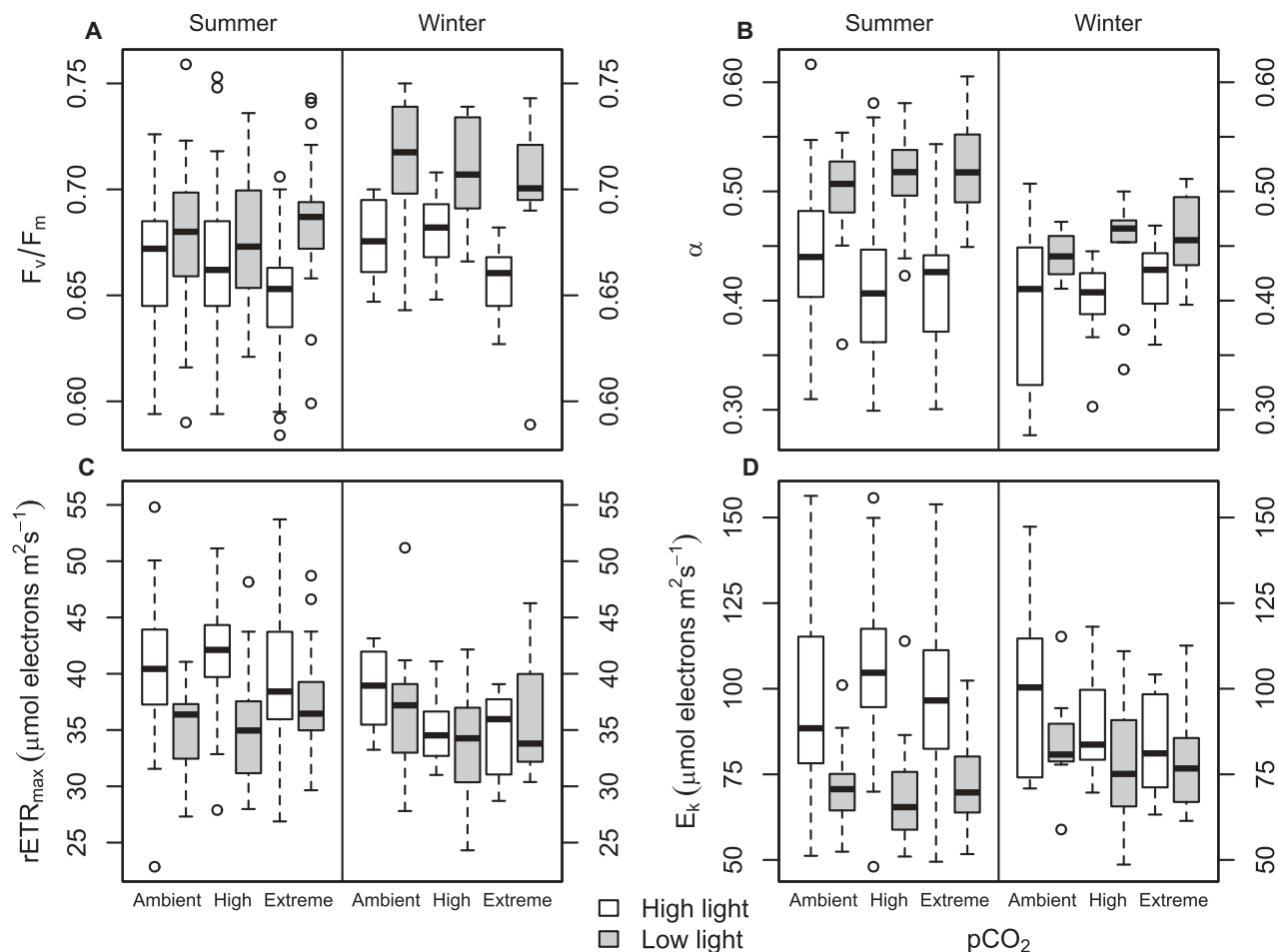


Fig. 3. Effects of pCO₂, light and season on: maximum potential quantum yield, F_v/F_m (A), light-limited photosynthetic efficiency, α (B), maximum relative electron transport rate, $rETR_{max}$ (C) and onset of light saturation, E_k (D) in *Fucus vesiculosus*.

summer (Table 4, Fig. 3D). pCO₂ treatment did not affect E_k .

Differences between the two seasons

Experiment timing had a substantial effect on all the measured parameters. The seasonal effect (winter-summer) was greater than that of 'high' pCO₂ treatment ($t = 3.81$, $df = 8.22$, $p < 0.021$), and the difference between 'extreme' pCO₂ treatment and season

was marginally significant ($t = 2.96$, $df = 9.97$, $p = 0.059$). We observed no significant difference between the relative magnitude (standardized main effects) of light treatment and season, nor pCO₂ treatments and light (Fig. 4, Supplementary table S2).

Discussion

Growth rate was not affected by either light or CO₂. Increasing CO₂ availability caused increasing carbon

Table 4. Statistical analyses (Mixed model or ANOVA) results for the chlorophyll fluorescence parameters measured.

	F_v/F_m		α		$rETR_{max}$		E_k	
	L	<i>p</i>	L	<i>p</i>	L	<i>p</i>	L	<i>p</i>
Summer								
pCO ₂ *Light	7.235	0.026*	7.079	0.029*	2.906	0.233	2.780	0.249
pCO ₂					0.567	0.753	0.539	0.763
Light					23.107	<0.001***	68.715	<0.001***
Winter								
pCO ₂ *Light	0.527	0.768	0.176	0.915	1.095	0.578	0.615	0.544
pCO ₂	8.812	0.012*	2.332	0.311	8.33	0.015*	4.415	0.125
Light	20.566	<0.001***	12.228	<0.001***	0.248	0.617	5.344	0.020*

Note: In the case of F_v/F_m and α , missing rows for single terms indicate that model selection ceased on a significant interaction term before single-term effects were analysed. Statistically significant ($p < 0.05$) effects are indicated in bold and with asterisks. Significance levels: *, $p < 0.05$, ***, $p < 0.001$. L indicates likelihood ratio (mixed model), and F an *F*-statistic (ANOVA), depending on analysis conducted. Degrees of freedom are the number of treatment levels (three for CO₂ and two for light) -1.

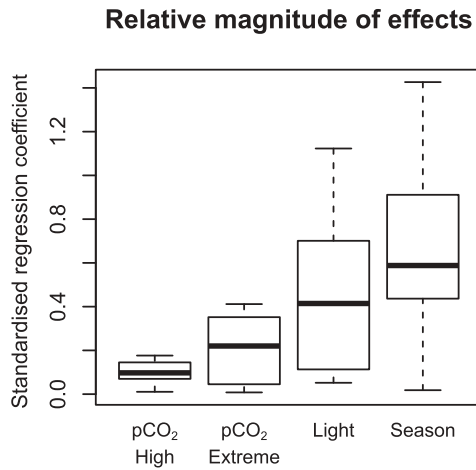


Fig. 4. Standardized regression coefficients of main effects indicating relative effect of each variable.

content in the algae during both seasons, but the effect was substantially stronger in winter. CO₂ increase also caused a decline in nitrogen content and an elevated C:N ratio, but only in winter. Light had a similar effect in the same direction. In chlorophyll *a* and light-use efficiency (F_v/F_m and α), these effects were in the same direction: increasing light decreased chlorophyll content in summer, and decreased light-use efficiency in both seasons. Increasing pCO₂ availability caused a decline in light-use efficiency, except in summer under high irradiance.

Effects of light and CO₂ on growth rate and carbon and nitrogen content

Although growth rate remained unaffected by pCO₂ or light, algae in high pCO₂ treatments were able to obtain an equal or greater share of inorganic carbon, especially during winter. As the sugar alcohol mannitol is the main carbon sink in *F. vesiculosus* (Bidwell & Ghosh, 1962; Lehvo *et al.*, 2001), an increase in inorganic carbon content may indicate increases in the mannitol pool. Although growth rates were not directly stimulated by pCO₂ treatments, an increase in mannitol storage could imply that high pCO₂ would have positive effects on growth rates in the longer term, because the energy stored in mannitol could later be utilized for growth (Lehvo *et al.*, 2001). In our study, increasing pCO₂ generally increased C content, which suggests that the alga could either elevate the carbon fixation rate or reduce respiratory losses of carbon under high pCO₂. Nitrogen content decreased under high pCO₂ in winter, and these together caused increases in the winter C:N ratio. In previous studies, OA (elevated pCO₂) has been observed to reduce nitrogen uptake in macroalgae by decreasing the activity of nitrate reductase (Hofmann *et al.*, 2013). Nitrogen

uptake is an active, energy-consuming process and thus its downregulation may arise if high pCO₂ reduces the energetic cost of carbon acquisition. The reason may be that when pCO₂ is elevated, plants need to invest less in photosynthetic energy capture, especially under abundant irradiance (as the decline in F_v/F_m and α indicate here), and thus have a reduced need for the upkeep of photosynthetic machinery as, in particular, the chlorophyll molecules are one of the main sinks of nitrogen in aquatic photoautotrophs (Kirk, 2011). Interestingly, pCO₂ treatment affected nitrogen content only during winter, but not in summer. *Fucus vesiculosus* takes up and stores nitrogen in winter, when seawater nitrogen content is high, to be utilized during rapid spring growth (Lehvo *et al.*, 2001). If exposure to high pCO₂ seawater alters nitrogen uptake and storage in the alga during winter, this may decrease the internal nitrogen pool size. In turn, this may potentially result in reduced growth during the main growth season in spring and summer, when seawater nitrogen concentrations are low, and internal nitrogen pools of the alga become depleted.

Changes in C:N ratio under OA have been observed in phytoplankton (Riebesell *et al.*, 2007) and macroalgae (Celis-Plá *et al.*, 2015). In our study, the pCO₂ treatments had significant effects on C:N ratios only during winter, not in summer, indicating that the response of *F. vesiculosus* carbon metabolism on OA may depend on season. High C:N ratios have been found to correlate with phenol synthesis in *F. vesiculosus* (Ilvessalo & Tuomi, 1989). Polyphenolic compounds, such as phlorotannins, are important defences against herbivory in *F. vesiculosus* (Koivikko *et al.*, 2005; Jormalainen & Ramsay, 2009). If phenol production is stimulated by elevated C:N ratios, this may also alter herbivory patterns in *F. vesiculosus*, thus potentially affecting the strength of trophic interactions in the littoral ecosystem. However, this remains unconfirmed in our study, as we did not measure algal phenol content. Our results, obtained with extreme pCO₂ treatments, contrast somewhat with those of Gutow *et al.* (2014), who observed declining C:N ratios and growth of *F. vesiculosus* under OA in a study conducted in a wholly marine ecosystem, albeit with ecologically more realistic pCO₂ treatments. It is possible that the responses in the relatively low-carbon Baltic and the Atlantic *F. vesiculosus* populations may be somewhat different. As our treatment levels for pCO₂ exceed those applied by Gutow *et al.* (2014), it is also possible that a threshold exists under which pCO₂ stimulates changes in the C:N ratio.

Effects on chlorophyll content and chlorophyll fluorescence parameters

Chlorophyll *a* content was only affected by light and only during summer. The low light-treated algae had higher chlorophyll *a* content, a pattern which has also been observed in the field (Rohde *et al.*, 2008). This is caused by increases in size and changes in the

number of photosynthetic units, especially in ratios of PS I to PS II, with PS I numbers declining in high irradiance (Kirk, 2011). pCO₂ treatment did not affect chlorophyll *a* content in either season. Although pCO₂ had no significant effect in winter, dropping pCO₂ from the model caused residual patterns in validation graphs when residuals were plotted against the pCO₂ treatment (Supplementary fig. S8), which indicates that dismissing the pCO₂ effect as non-significant should be considered with caution.

Our study identified interacting effects between pCO₂ and light, with declining light-use efficiency (decrease in F_v/F_m and α) emerging under high pCO₂ under high irradiances. Under low irradiance, α and F_v/F_m remained high in both seasons, probably because the plants had to maintain relatively high light-harvesting capacities to supply dark reactions with sufficient reductant levels for carbon fixation. Under high irradiance, conversely, light-use efficiency generally declined, especially during summer, which was expected, as there was no pressure for efficient light utilization under high irradiance. However, the decline in F_v/F_m and α under high pCO₂ and high irradiance suggests that the light energy used for carbon fixation also declined under high pCO₂. This supports the idea that the algae could have down-regulated CCM usage and supplied their carbon fixation increasingly with free CO₂, but as we did not analyse carbon isotope ratios in algal tissue this remains speculative. In eelgrass *Zostera marina*, increasing carbon (CO₂) availability reduces photosynthetic light requirements and allows the plants to sustain their growth with lower accumulated daily irradiance (Zimmerman *et al.*, 1997). Saturation of irradiance requirement, a similar process, is a plausible explanation for the decline in F_v/F_m and α under high pCO₂ and high irradiance observed here.

In conclusion, the observed responses to experimental treatments showed strong differences between the two time points sampled, which suggest seasonal variation in many physiological traits of *F. vesiculosus*, especially nitrogen and carbon content and growth rate (Lehvo *et al.*, 2001). Responses to pCO₂ treatments were rather small compared with seasonal changes in all parameters measured. As CO₂ in our experiment was administered by manipulating the pH of seawater, the changing alkalinity, salinity and temperature between the two experiments caused the pCO₂ levels to vary between corresponding pH treatments in summer and winter (Supplementary fig. S4), with the winter experiment having higher pCO₂ levels for each corresponding pCO₂ treatment level when compared with summer (Table 1). As temperature strongly affects the physiology of *F. vesiculosus* (Kraufvelin *et al.*, 2012; Al-Janabi *et al.*, 2016a), differences in seawater temperature between the two

seasons may partly explain the observed dissimilarities. It is also noteworthy that with two 3-week experiments, the evaluation of seasonal responses relies on two snapshots in time and thus may not provide a thorough picture of seasonal changes in the physiology of *F. vesiculosus*, although, with C and N content in particular our results are in general agreement with previous studies conducted in the same region (Lehvo *et al.*, 2001). The responses observed in such relatively short experiments may also differ from the effects of longer exposure to the variables tested. Also, if the duration of CO₂ exposure is short, not all of the effects observed here may be realized *in situ*, as alterations in carbon physiology in macroalgae may occur over a time course of days to weeks (Hofmann *et al.*, 2013), whereas alterations in irradiance conditions may cause physiological responses in minutes or hours (Falkowski & Raven, 2007; Kirk, 2011).

In summary, CO₂ and light had substantial effects on *F. vesiculosus* physiology, most notably in C and N content and photosynthetic parameters. The directions of pCO₂ and light effects were often in the same direction. The physiology of *F. vesiculosus* shows strong seasonality in the northern Baltic, and the potential effects of high pCO₂ vary in time. However, compared with the seasonal changes in all measured parameters, the effects of CO₂ were small in magnitude, at least on the mature vegetative thalli investigated in this study. In contrast, light availability had a substantial effect on many of the parameters measured. Light and pCO₂ had similar effects (in the same direction) on carbon and nitrogen content, especially in winter, which suggests that during the winter experiment the algae were light- and carbon-limited, possibly because the internal carbon pools had been depleted by energetic requirements to sustain growth.

In our study, *F. vesiculosus* growth rates were not affected by either pCO₂ or light treatments. As growth rate is dependent on a multitude of physiological processes, we consider it to be the strongest performance proxy. Based on the results of our study, we conclude that stochastic upwelling of pCO₂-rich water may potentially have some beneficial effects (such as increased carbon accumulation in winter), but the full effects on algal fitness appear rather small. Consequently, CO₂ enrichment from other sources, such as anthropogenic ocean acidification, probably has only minor effects on *F. vesiculosus*.

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No potential conflict of interest was reported by the authors.

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Supplementary information

The following supplementary material is accessible via the Supplementary Content tab on the article's online page at <http://10.1080/09670262.2019.1572226>

Supplementary table S1. Variables for which variance structures were applied in the analysis.

Supplementary table S2. Results of Games-Howell *post hoc* test of the differences of standardized main effects of different factors (Season, Light, pCO₂ high, PCO₂ extreme).

Supplementary fig. S1. Salinity and temperature during the two experiments.

Supplementary fig. S2. pH levels during the two experiments for the two light treatments.

Supplementary fig. S3. Titrated and calculated alkalinity values during the winter experiment (day 14) for the three pCO₂ treatments.

Supplementary fig. S4. pCO₂, CO₃, HCO₃⁻ and total dissolved inorganic carbon (DIC) in the seawater during both experiments.

Supplementary figure S5. pH and temperature and calculated pCO₂ and DIC values at five coastal sites near TZS with dense *Fucus* vegetation measured in 2017.

Supplementary fig. S6. Map of HELCOM monitoring stations from which data were downloaded, and Tvärminne Zoological Station.

Supplementary fig S7. Values of temperature, pH, alkalinity, DIC, pCO₂ and HCO₃⁻ from the HELCOM monitoring stations (Fig. S6).

Supplementary fig. S8. Normalized residuals of final model (pCO₂ and Light dropped) in chl *a* analysis (winter experiment) plotted against pCO₂ treatment.

Supplementary fig. S9. Residuals of two candidate models for Fv/Fm analysis.

Supplementary fig. S10. Model validation graphs for the full model for alpha in the summer experiment (pCO₂*light interaction and various variances allowed for both levels of light).

Supplementary fig. S11. Model validation graphs for a no-interaction model for alpha in the summer experiment (pCO₂ and light, interaction dropped and various variances allowed for both levels of light).

Author contributions

A. Takolander: original idea, designing the study, conducting the experiment, data analysis, writing the manuscript;

E. Leskinen: designing the study, writing the manuscript; M. Cabeza: designing the study, writing the manuscript.

ORCID

Antti Takolander  <http://orcid.org/0000-0002-9237-3765>

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References

- Al-Janabi, B., Kruse, I., Graiff, A., Karsten, U. & Wahl, M. (2016a). Genotypic variation influences tolerance to warming and acidification of early life-stage *Fucus vesiculosus* L. (Phaeophyceae) in a seasonally fluctuating environment. *Marine Biology*, **163**: 14.
- Al-Janabi, B., Kruse, I., Graiff, A., Winde, V., Lenz, M. & Wahl, M. (2016b). Buffering and amplifying interactions among OAW (Ocean Acidification & Warming) and nutrient enrichment on early life-stage *Fucus vesiculosus* L. (Phaeophyceae) and their carry over effects to hypoxia impact. *PLoS ONE*, **11**: e0152948.
- Bidwell, R.G.S. & Ghosh, N.R. (1962). Photosynthesis and metabolism in marine algae: IV. The fate of C14-mannitol in *Fucus vesiculosus*. *Canadian Journal of Botany*, **40**: 803–807.
- Bonsdorff, E., Blomqvist, E.M., Mattila, J. & Norkko, A. (1997). Coastal eutrophication: causes, consequences and perspectives in the Archipelago areas of the northern Baltic Sea. *Estuarine, Coastal and Shelf Science*, **44**: 63–72.
- Borges, A.V., Schiettecatte, L.-S., Abril, G., Delille, B. & Gazeau, F. (2006). Carbon dioxide in European coastal waters. *Estuarine, Coastal and Shelf Science*, **70**: 375–87.
- Burnett, L.E. (1997). The challenges of living in hypoxic and hypercapnic aquatic environments. *American Zoologist*, **37**: 633–40.
- Carstensen, J., Andersen, J.H., Gustafsson, B.G. & Conley, D.J. (2014). Deoxygenation of the Baltic Sea during the last century. *Proceedings of the National Academy of Sciences USA*, **111**: 5628–33.
- Celis-Plá, P.S.M., Hall-Spencer, J.M., Horta, P.A., Milazzo, M., Korbee, N., Cornwall, C.E. & Figueroa, F.L. (2015). Macroalgal responses to ocean acidification depend on nutrient and light levels. *Frontiers in Marine Science*, **2**.
- Conley, D.J., Carstensen, J., Aigars, J., Axe, P., Bonsdorff, E., Eremina, T.R., Haahti, B., Humborg, C., Jonsson, P., Kotta, J., Lännegren, C., Larsson, U., Maximov, A., Medina, M.R., Lysiak-Pastuszek, E., Remeikaitė-Nikienė, N., Walve, J., Wilhelms, S. & Zillén, L. (2011). Hypoxia is increasing in the coastal zone of the Baltic Sea. *Environmental Science and Technology*, **45**: 6777–6783.
- Cornwall, C.E., Hepburn, C.D., Pritchard, D., Currie, K.I., McGraw, C.M., Hunter, K. A. & Hurd, C.L. (2012). Carbon-use strategies in macroalgae: differential responses to lowered pH and implications for ocean acidification. *Journal of Phycology*, **48**: 137–144.
- Fabry, V., Seibel, B., Feely, R. & Orr, J. (2008). Impacts of ocean acidification on marine fauna and ecosystem processes. *ICES Journal of Marine Sciences*, **65**: 414–32.
- Falkowski, P. & Raven, J. (2007). *Aquatic Photosynthesis*. 2nd ed. Princeton University Press, Princeton.

- Gao, K., Xu, J., Gao, G., Li, Y. & Hutchins, D. (2012). Rising CO₂ and increased light exposure synergistically reduce marine primary productivity. *Nature Climate Change*, **2**: 519–523.
- Gattuso, J.-P., Epitalon, J.-M. & Lavigne, H. (2015). sea-carb: Seawater Carbonate Chemistry.
- Genty, B., Briantais, J. & Baker, N. (1989). The relationship between the quantum yield of photosynthetic electron transport and quenching of chlorophyll fluorescence. *Biochimica et Biophysica Acta – General Subjects*, **990**: 87–92.
- Giordano, M. & Maberly, S.C. (1989). Distribution of carbonic anhydrase in British marine macroalgae. *Oecologia*, **81**: 534–539.
- Gutow, L., Rahman, M.M., Bartl, K., Saborowski, R., Bartsch, I. & Wiencke, C. (2014). Ocean acidification affects growth but not nutritional quality of the seaweed *Fucus vesiculosus* (Phaeophyceae, Fucales). *Journal of Experimental Marine Biology and Ecology*, **453**: 84–90.
- Haapala, J. (1994). Upwelling and its influence on nutrient concentration in the coastal area of the Hanko Peninsula, entrance of the Gulf of Finland. *Estuarine, Coastal and Shelf Science*, **38**: 507–521.
- Hofmann, L.C., Straub, S. & Bischof, K. (2013). Elevated CO₂ levels affect the activity of nitrate reductase and carbonic anhydrase in the calcifying rhodophyte *Corallina officinalis*. *Journal of Experimental Botany*, **64**: 899–908.
- ICES (2014). ICES Dataset on Ocean Hydrography. The International Council for the Exploration of the Sea, Copenhagen.
- Ilvessalo, H. & Tuomi, J. (1989). Nutrient availability and accumulation of phenolic compounds in the brown alga *Fucus vesiculosus*. *Marine Biology*, **101**: 115–119.
- Jormalainen, V. & Ramsay, T. (2009). Resistance of the brown alga *Fucus vesiculosus* to herbivory. *Oikos*, **118**: 713–722.
- Kautsky, H., Kautsky, L., Kautsky, N., Kautsky, U. & Lindblad, C. (1992). Studies on the *Fucus vesiculosus* community in the Baltic Sea. *Acta Phytogeographica Suecica*, **78**: 33–48.
- Kirk, J.T.O. (2011). *Light and Photosynthesis in Aquatic Environments*. 3rd ed. Cambridge University Press, Cambridge.
- Koch, M., Bowes, G., Ross, C. & Zhang, X.-H. (2013). Climate change and ocean acidification effects on seagrasses and marine macroalgae. *Global Change Biology*, **19**: 103–132.
- Koivikko, R., Lopenen, J., Honkanen, T. & Jormalainen, V. (2005). Contents of soluble, cell-wall-bound and exuded phlorotannins in the brown alga *Fucus vesiculosus*, with implications on their ecological functions. *Journal of Chemical Ecology*, **31**: 195–212.
- Koroleff, F. (1983a). Simultaneous oxidation of nitrogen and phosphorus compounds by persulfate. *Methods of Seawater Analysis*, **2**: 205–206.
- Koroleff, F. (1983b). Total and organic nitrogen. *Methods of Seawater Analysis*, **2**: 162–173.
- Kraufvelin, P., Ruuskanen, A.T., Bäck, S. & Russell, G. (2012). Increased seawater temperature and light during early springs accelerate receptacle growth of *Fucus vesiculosus* in the northern Baltic proper. *Marine Biology*, **159**: 1795–1807.
- Kübler, J.E. & Dudgeon, S.R. (2015). Predicting effects of ocean acidification and warming on algae lacking carbon concentrating mechanisms. *PLoS ONE*, **10**: e0132806.
- Lehvo, A., Bäck, S. & Kiirikki, M. (2001). Growth of *Fucus vesiculosus* L. (Phaeophyta) in the Northern Baltic Proper: energy and nitrogen storage in seasonal environment. *Botanica Marina*, **44**: 345–350.
- Leivuori, M., Koivikko, R., Sarkkinen, M., Tervonen, K., Lanteri, S., Väisänen, R. & Ilmakunnas, M. (2018). Interlaboratory Proficiency Test 03/2018.
- Lindström, M. (2000). Seasonal changes in the underwater light milieu in a Finnish Baltic Sea coastal locality. *Geophysica*, **36**: 215–232.
- Liu, Y., Xu, J. & Gao, K. (2012). CO₂-driven seawater acidification increases photochemical stress in a green alga. *Phycologia*, **51**: 562–566.
- Long, M.H., Rheuban, J.E., Berg, P. & Ziemann, J.C. (2012). A comparison and correction of light intensity loggers to photosynthetically active radiation sensors. *Limnology and Oceanography: Methods*, **10**: 416–424.
- Lüning, K. (1990). *Seaweeds: Their Environment, Biogeography and Ecophysiology*. Wiley-Interscience Publication, New York.
- Maxwell, K. & Johnson, G.N. (2000). Chlorophyll fluorescence – a practical guide. *Journal of Experimental Botany*, **51**: 659–668.
- Meier, H.E.M., Hordoir, R., Andersson, H.C., Dieterich, C., Eilola, K., Gustafsson, B.G., Höglund, A. & Schimanke, S. (2012). Modeling the combined impact of changing climate and changing nutrient loads on the Baltic Sea environment in an ensemble of transient simulations for 1961–2099. *Climate Dynamics*, **39**: 2421–2441.
- Melzner, F., Thomsen, J., Koeve, W., Oschlies, A., Gutowska, M.A., Bange, H.W., Hansen, H.P. & Körtzinger, A. (2013). Future ocean acidification will be amplified by hypoxia in coastal habitats. *Marine Biology*, **160**: 1875–1888.
- Mercado, J.M. & Gordillo, F.J.L. (2011). Inorganic carbon acquisition in algal communities: are the laboratory data relevant to the natural ecosystems? *Photosynthetic Research*, **109**: 257–267.
- Middelboe, A. & Hansen, P. (2007). High pH in shallow-water macroalgal habitats. *Marine Ecology Progress Series*, **338**: 107–117.
- Müller, J.D., Schneider, B. & Rehder, G. (2016). Long-term alkalinity trends in the Baltic Sea and their implications for CO₂-induced acidification. *Limnology and Oceanography*, **61**: 1984–2002.
- Neumann, T., Eilola, K., Gustafsson, B., Müller-Karulis, B., Kuznetsov, I., Meier, H.E.M. & Savchuk, O.P. (2012). Extremes of temperature, oxygen and blooms in the Baltic sea in a changing climate. *Ambio*, **41**: 574–585.
- Olischläger, M., Bartsch, I., Gutow, L. & Wiencke, C. (2013). Effects of ocean acidification on growth and physiology of *Ulva lactuca* (Chlorophyta) in a rock-pool-scenario. *Phycological Research*, **61**: 180–190.
- Omstedt, A., Gustafsson, E. & Wesslander, K. (2009). Modelling the uptake and release of carbon dioxide in the Baltic Sea surface water. *Continental Shelf Research*, **29**: 870–885.
- Omstedt, A., Edman, M., Anderson, L.G. & Laudon, H. (2010). Factors influencing the acid-base (pH) balance in the Baltic Sea: a sensitivity analysis. *Tellus B: Chemical and Physical Meteorology*, **62**: 280–295.
- Omstedt, A., Edman, M., Claremar, B., Frodin, P., Gustafsson, E., Humborg, C., Hägg, H. & Mörth, M., Rutgersson, A., Schurgers, G., Smith, B., Wällstedt, T. & Yurova, A. (2012). Future changes in the Baltic Sea acid-base (pH) and oxygen balances. *Tellus B: Chemical and Physical Meteorology*, **64**.
- Orr, J.C., Fabry, V.J., Aumont, O., Bopp, L., Doney, S.C., Feely, R. A., Gnanadesikan, A., Gruber, N., Ishida, A., Joos, F., Key, R.M., Lindsay, K., Maier-Reimer, E., Matear, R.J., Monfray, P., Mouchet, A., Najjar, R.G.,

- Plattner, G.-K., Rodgers, K.B., Sabine, C.L., Sarmiento, J. L., Schlitzer, R., Slater, R.D., Totterdell, I.J., Weirig, M.-F., Yamanaka, Y. & Yool, A. (2005). Anthropogenic ocean acidification over the twenty-first century and its impact on calcifying organisms. *Nature*, **437**: 681–686.
- Pajusalu, L., Martin, G., Põllumäe, A. & Paalme, T. (2013). Results of laboratory and field experiments of the direct effect of increasing CO₂ on net primary production of macroalgal species in brackish-water ecosystems. *Proceedings of the Estonian Academy of Sciences*, **62**: 148.
- Pinheiro, J., Bates, D., DebRoy, S. & R Core Team (2015). nlme: Linear and Nonlinear Mixed Effects Models. <https://CRAN.R-project.org/package=nlme>.
- Platt, T., Gallegos, C. & Harrison, W. (1981). Photoinhibition of photosynthesis in natural assemblages of marine phytoplankton. *Journal of Marine Research*, **38**: 687–701.
- R Core Team (2015). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna.
- Ralph, P.J. & Gademann, R. (2005). Rapid light curves: a powerful tool to assess photosynthetic activity. *Aquatic Botany*, **82**: 222–237.
- Raven, J.A. & Hurd, C.L. (2012). Ecophysiology of photosynthesis in macroalgae. *Photosynthetic Research*, **113**: 105–125.
- Raven, J.A. & Samuelsson, G. (1988). Ecophysiology of *Fucus vesiculosus* L. close to its northern limit in the Gulf of Bothnia. *Botanica Marina*, **31**: 399–410.
- Raven, J. A., Giordano, M., Beardall, J. & Maberly, S.C. (2011). Algal and aquatic plant carbon concentrating mechanisms in relation to environmental change. *Photosynthetic Research*, **109**: 281–296.
- Raven, J.A., Beardall, J. & Giordano, M. (2014). Energy costs of carbon dioxide concentrating mechanisms in aquatic organisms. *Photosynthetic Research*, **121**: 111–124.
- Riebesell, U., Schulz, K.G., Bellerby, R.G.J., Botros, M., Fritsche, P., Meyerhöfer, M., Neill, C., Nondal, G., Oschlies, A., Wohlers, J. & Zöllner, E. (2007). Enhanced biological carbon consumption in a high CO₂ ocean. *Nature*, **450**: 545–548.
- Ritchie, R. (2008). Universal chlorophyll equations for estimating chlorophylls *a*, *b*, *c*, and *d* and total chlorophylls in natural assemblages of photosynthetic organisms using acetone, methanol or ethanol solvents. *Photosynthetica*, **46**: 115–126.
- Rohde, S., Hiebenthal, C., Wahl, M., Karez, R. & Bischof, K. (2008). Decreased depth distribution of *Fucus vesiculosus* (Phaeophyceae) in the Western Baltic: effects of light deficiency and epibionts on growth and photosynthesis. *European Journal of Phycology*, **43**: 143–150.
- Rugiu, L., Manninen, I., Rothäusler, E. & Jormalainen, V. (2018). Tolerance and potential for adaptation of a Baltic Sea rockweed under predicted climate change conditions. *Marine Environmental Research*, **134**: 76–84.
- Saderne, V., Fietzek, P. & Herman, P.M.J. (2013). Extreme variations of pCO₂ and pH in a macrophyte meadow of the Baltic Sea in summer: evidence of the effect of photosynthesis and local upwelling. *PLoS ONE*, **8**: e62689.
- Salonen, K. (1981). Rapid and precise determination of total inorganic carbon and some gases in aqueous solutions. *Water Research*, **15**: 403–406.
- Schagerström, E., Forslund, H., Kautsky, L., Pärnoja, M. & Kotta, J. (2014). Does thalli complexity and biomass affect the associated flora and fauna of two co-occurring *Fucus* species in the Baltic Sea? *Estuarine, Coastal and Shelf Science*, **149**: 187–193.
- SFS 3005 (1981). Veden alkaliteetin ja asiditeetin määrittäminen. Potentiometrinen titraus. Finnish Standards Association. 5 pp.
- Silsbe, G.M. & Malkin, S.M. (2015). phytotools: Phytoplankton Production Tools. R package.
- Snøeijs-Leijonmalm, P., Schubert, H. & Radziejewska, T. (2017). *Biological Oceanography of the Baltic Sea*. Springer Science & Business Media, Amsterdam.
- Sunda, W.G. & Cai, W.-J. (2012). Eutrophication induced CO₂-acidification of subsurface coastal waters: interactive effects of temperature, salinity, and atmospheric pCO₂. *Environmental Science and Technology*, **46**: 10651–10659.
- Surif, M.B. & Raven, J.A. (1989). Exogenous inorganic carbon sources for photosynthesis in seawater by members of the Fucales and the Laminariales (Phaeophyta): ecological and taxonomic implications. *Oecologia*, **78**: 97–105.
- Takolander, A., Cabeza, M. & Leskinen, E. (2017a). Climate change can cause complex responses in Baltic Sea macroalgae: a systematic review. *Journal of Sea Research*, **123**: 16–29.
- Takolander, A., Leskinen, E. & Cabeza, M. (2017b). Synergistic effects of extreme temperature and low salinity on foundational macroalga *Fucus vesiculosus* in the northern Baltic Sea. *Journal of Experimental Marine Biology and Ecology*, **495**: 110–118.
- Thomas, H. & Schneider, B. (1999). The seasonal cycle of carbon dioxide in Baltic Sea surface waters. *Journal of Marine Systems*, **22**: 53–67.
- van Helmond, N.A.G.M., Jilbert, T. & Slomp, C.P. (2018). Hypoxia in the Holocene Baltic Sea: comparing modern versus past intervals using sedimentary trace metals. *Chemical Geology*, **493**: 478–490.
- Verspagen, J.M.H., Van de Waal, D.B., Finke, J.F., Visser, P. M. & Huisman, J. (2014). Contrasting effects of rising CO₂ on primary production and ecological stoichiometry at different nutrient levels. *Ecology Letters*, **17**: 951–960.
- Vuorinen, I., Hänninen, J., Rajasilta, M., Laine, P., Eklund, J., Montesino-Pouzols, F., Corona, F., Junker, K., Meier, H.E. M. & Dippner, J.W. (2015). Scenario simulations of future salinity and ecological consequences in the Baltic Sea and adjacent North Sea areas – implications for environmental monitoring. *Ecological Indicators*, **50**: 196–205.
- Waern, M. (1952). *Rocky-shore Algae in the Öregrund Archipelago*. Almqvist & Wiksells Boktr., Uppsala.
- Wahl, M., Jormalainen, V., Eriksson, B.K., Coyer, J.A., Molis, M., Schubert, H., Dethier, M., Karez, R., Kruze, I., Lenz, M., Pearson, G., Rhode, S., Wikström, S. A. & Olsen, J. L. (2011). Stress ecology in *Fucus*: abiotic, biotic and genetic interactions. *Advances in Marine Biology*, **59**: 37–105.
- Werner, F.J., Graiff, A. & Matthiessen, B. (2016). Temperature effects on seaweed-sustaining top-down control vary with season. *Oecologia*, **180**: 889–901.
- Wu, H., Zou, D. & Gao, K. (2008). Impacts of increased atmospheric CO₂ concentration on photosynthesis and growth of micro- and macro-algae. *Science China Life Sciences*, **51**: 1144–1150.
- Zimmerman, R.C., Kohrs, D.G., Steller, D.L. & Alberte, R.S. (1997). Impacts of CO₂ enrichment on productivity and light requirements of eelgrass. *Plant Physiology*, **115**: 599–607.
- Zuur, A.F., Ieno, E.N., Walker, N.J., Saveliev, A.A. & Smith, G.M. (2009). *Mixed Effects Models and Extensions in Ecology with R*. Springer, New York.